Prodigy Anesthesia Quick Review

Caudal Block

What is a caudal epidural?	A caudal epidural is an epidural anesthetic administered via the sacral hiatus that acts on the sacral dermatomes. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097.*
For what procedures is a caudal anesthetic indicated?	Caudal anesthesia is used most commonly as an adjunct to general anesthesia in pediatric surgery. It can also be used for perirectal surgery, urologic surgery, lower extremity orthopedic surgery, labor and delivery, and chronic pain. Clubfoot repair, inguinal herniorrhaphy, and circumcision are also surgeries that would benefit from the pain relief provided by a caudal block. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097.*
What is the traditional technique for performing a caudal anesthetic?	The patient is positioned prone or lateral. The sacral hiatus is identified by palpation and a 22- to 25-gauge needle is inserted in the midline between the two sacral cornua. Some clinicians advocate using a 20-gauge IV catheter in place of a standard needle. A popping sensation can be felt as the needle passes through the sacrococygeal ligament. The angle of the needle is lowered parallel to the sacrum and the needle is advanced 1-3 cm into the epidural space. Gentle aspiration is performed to detect inadvertant entry in the subarachnoid space or a blood vessel. The local anesthetic is injected incrementally while the opposite hand palpates the skin over the sacrum for bulging indicative of subcutaneous injection. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
What are the anatomic landmarks that can be used to identify the correct placement of the needle for a caudal block?	The two sacral cornua, the coccyx, and the posterior superior iliac spines. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1197.*
In what position is the patient typically placed for a caudal anesthetic?	The patient can be placed in either the prone or lateral position. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
How is the volume of local anesthetic estimated for an adult undergoing caudal anesthesia?	A 3 mL test dose is administered first. For sacral procedures, 12-15 mL is sufficient. 20-30 mL is usually required for anesthesia up to a T10 dermatome. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*

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How is the appropriate volume of local anesthetic calculated for a caudal anesthetic for a pediatric patient?	0.5 to 1 mL of solution per kg of bodyweight should be injected to achieve an umbilical level of anesthesia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
What is the 'whoosh' test used during caudal anesthesia?	The 'whoosh' test is performed to determine correct placement of the epidural needle. A stethoscope is placed over the lumbar spine and 2-3 mLs of air is injected through the epidural needle. If the needle is correctly placed, a 'whoosh' sound should be heard. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 61.*
If a catheter is to be inserted for a continuous caudal epidural anesthetic, what needle is usually used and why?	A Crawford needle is used because it has a bevel that faces forward which more closely aligns with the direction the catheter needs to travel in the caudal epidural space. The Tuohy needle typically used in lumbar epidural anesthesia faces upward toward the wall of the caudal epidural canal. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 60.*
What ligament overlies the sacral hiatus?	The sacrococcygeal ligament *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 61.*
You are performing a caudal anesthetic on an awake patient. As you begin to inject the local anesthetic, the patient complains of paresthesia and a feeling of 'fullness' in the sacral area. What should you do next?	These are typical sensations when performing a caudal epidural correctly. You should reassure the patient that this is normal and continue incremental injection. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 61.*
What is the dose of clonidine that should be added to the local anesthetic to augment a caudal anesthetic?	1 mcg/kg of bodyweight *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
What are the advantages of using clonidine rather than opioids to augment a caudal anesthetic?	Clonidine is comparable to opioids in its ability to augment a caudal epidural block without the accompanying respiratory depression, nausea, and delayed recovery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
Why is caudal anesthesia used predominantly in pediatric patients?	After the age of 12, the sacral bone structure begins to change. The sacral hiatus is more difficult to locate and the spread of local anesthetic in the caudal epidural space is less reliable. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097-1098.*

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When during the perioperative period is a caudal anesthetic usually performed in a pediatric patient?	It is usually performed after induction of general anesthesia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*
Is an inadvertant intrathecal injection more common with a caudal epidural performed on a child or an adult? Why?	The epidural sac extends to S1 in adults and S3 in children making inadvertant intrathecal injection more likely in children. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 964.*
What psychological effect of a caudal anesthetic should you consider prior to performing the procedure on a toddler?	Although not an issue in infants who cannot walk, the loss of motor innervation and inability to walk may frighten a toddler. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1098.*

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